



SALAS O'BRIEN

3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Lake Preserve Jackson Trail; LGI Homes
Location: Hoschton, GA
Description: Savannah - A/B, Slab, CBonus

Date: August 25, 2024
By: JTH / TJH
House Faces: Worst NorthEast

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	2,096 ft ²
Volume	18,744 ft ³
Percentage Glass	9 %
Number of Bedrooms	3

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	8448
SEER	14
ACH50	5.0
Mechanical Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x6	R-13 batt	
	W2	0.143	Basement	Concrete + 2x6	R-5	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.2	Exterior Door		R-5	
	D2	0.5	Garage Door		R-2	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling	2x4	R-38 blown	
	R2	0.026	Vaulted	2x4	R-38 blown	
Floors	#	U-Value	Description/Notes			
	F1	0.053	Framed		R-19 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.32	Blinds/Draperies	SHGC 0.22
	G2					
	G3					

LOAD SUMMARY					
		Unit 1 - Whole House			
		Cooling	Heating		
Sensible Load		24781 Btu/hr	27633 Btu/hr		
Minimum Nominal Capacity		2.75 Tons	8.10 kW		
Load Density		761 SF/Ton	3.9 W/SF		
Floor Area Served		2096.0 SF	2096.0 SF		
Mechanical Ventilation		N/A CFM	N/A CFM		
System Size		3.0 Tons	10.0 kW		
Leaving Air Temperature		57.2 deg F	96.3 deg F		
Installed Capacity		699 SF/Ton	4.8 W/SF		

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

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[illegible]

Unit 1

Size

3.00

Tons

Airflow

1200

CFM

Heat kW

10

kW

First Floor Airflows

Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Foyer	32	50	41	
Office	71	103	87	
Pantry	7	15	11	
Kitchen	159	73	116	
Garage Entry Hall	33	46	39	
Laundry	70	47	59	
Primary WIC	30	53	42	
Primary Bath	42	79	61	
Primary Bed	105	108	106	
Powder	6	9	7	
Great Rm	180	151	165	
Dining	32	56	44	
-	-	-	-	
-	-	-	-	

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DX Cooling kWh	2112
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RC Heating kWh	3384
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Level(s): First Floor

Total kWh	5496
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SEER 14

COOLING LOAD CALCULATION						Ceiling Height	9		9		9		9		9		9		9		9		9		9												
Components					CLTD (°F)	U-Value (Btu/hr-sq-ft-F)	Cooling Load Factor	Entire House		Foyer		Office		Pantry		Kitchen		Garage Entry Hall		Laundry		Primary WIC		Primary Bath		Primary Bed		Powder		Great Rm		Dining					
Type	Description			Qty				Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	
Glass	G1	NorthWest	Right	(E)	18	Btu/hr-sf	18	325	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	12	217	6	108	-	-	-	-	
	G1	North		(SE)	6	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	NorthEast	Front	(S)	7	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	East		(SW)	13	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	SouthEast	Left	(W)	14	Btu/hr-sf	10	139	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	South		(NW)	15	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	SouthWest	Back	(N)	26	Btu/hr-sf	73	1895	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	West		(NE)	29	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Door	G1	Sky Light			45	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	Shaded			6	Btu/hr-sf	46	270	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Above Grade Walls	D1	Door	21	0.20	4.20	Btu/hr-sf	14	61	10	59	36	211	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	D2	Door	30	0.50	15.00	Btu/hr-sf	18	267	10	42	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthWest	19	0.08	1.46	Btu/hr-sf	467	683	-	-	113	166	37	54	127	185	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	North	13	0.08	1.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthEast	19	0.08	1.46	Btu/hr-sf	98	144	29	43	69	101	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	East	23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthEast	21	0.08	1.62	Btu/hr-sf	334	540	-	-	25	40	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-				
	W1	South	16	0.08	1.23	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest	21	0.08	1.62	Btu/hr																															

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